

Bootstrap-Based Assessment of Uncertainty in Logistic Regression for Breast Cancer Diagnosis

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1. Abstract

This project investigates the stability and reliability of logistic regression estimates for predicting breast cancer malignancy using features from the Wisconsin Diagnostic Breast Cancer dataset. Although logistic regression is widely used in biomedical applications, its standard analytical inference relies on large-sample approximations that may underestimate uncertainty in complex or highly correlated datasets. To evaluate the robustness of model-based inference, I implemented three bootstrap resampling procedures—nonparametric, parametric, and residual-based bootstrapping—and generated 2,000 bootstrap replicates for each method. From these distributions, I computed bootstrap standard errors, bias estimates, and percentile confidence intervals, and compared them with the traditional Wald standard errors and confidence intervals produced by `glm()` in R.

The bootstrap distributions revealed that *mean_texture*, *mean_smoothness*, and *mean_concave_points* were consistently strong and stable predictors of malignancy, with bootstrap confidence intervals that remained entirely above zero across all resampling methods. In contrast, predictors such as *mean_radius*, *mean_perimeter*, *mean_compactness*, *mean_concavity*, *mean_symmetry*, and *mean_fractal_dimension* showed substantial variability and confidence intervals that overlapped zero, indicating weak or unstable effects once multicollinearity across features was accounted for. Comparisons between bootstrap-based and analytical uncertainty estimates showed close agreement, with bootstrap standard errors mirroring the `glm` Wald standard errors and no evidence that traditional inference systematically underestimated variability.

Overall, the bootstrap results confirm the reliability of logistic regression inference for this dataset and provide a clear assessment of which predictors carry robust diagnostic signal. This analysis demonstrates the value of bootstrap methods for validating model-based uncertainty estimates and for identifying stable, reproducible predictors in biomedical classification problems.

2. Introduction

Breast cancer diagnosis is a critical area of biomedical research where statistical modeling plays an essential role in distinguishing malignant from benign tumors based on quantitative features extracted from medical imaging. The Wisconsin Diagnostic Breast Cancer dataset provides several such predictors—such as tumor radius, texture, smoothness, and concavity—that are commonly used in classification models. Logistic regression is one of the most widely applied methods for modeling binary clinical outcomes because of its interpretability and well-established theoretical properties. However, logistic regression inference traditionally relies on large-sample approximations, which may underestimate uncertainty when predictors are highly correlated or when the underlying assumptions of the model are strained. Bootstrapping offers a flexible, data-driven approach to assessing parameter variability without relying on asymptotic

theory, making it especially valuable in biomedical settings where predictor relationships may be complex. In this project, I apply nonparametric, parametric, and residual-based bootstrap procedures to evaluate the stability of logistic regression coefficients, quantify their uncertainty, and determine which predictors contribute consistently to accurate malignant tumor classification. By comparing bootstrap-based uncertainty estimates to the standard Wald intervals produced by `glm()` in R, this analysis provides insight into the reliability of logistic regression inference and highlights the benefits of resampling methods for assessing predictive robustness.

3. Literature review

Breast cancer classification has long been a focus of biomedical statistics and machine learning research, with numerous studies emphasizing the importance of reliable predictive modeling for early detection and improved clinical outcomes. The Wisconsin Diagnostic Breast Cancer (WDBC) dataset in particular has become a widely used benchmark for evaluating statistical and computational classification methods (Wolberg et al., 1995). Early work demonstrated the utility of logistic regression as a transparent and interpretable model for distinguishing malignant from benign tumors using morphometric features extracted from digitized images of fine-needle aspirates (Street et al., 1993). Because logistic regression provides interpretable coefficients and clinically meaningful odds ratios, it remains a preferred modeling framework in medical research despite the increasing adoption of more complex machine learning algorithms.

A significant challenge in logistic regression arises from its reliance on asymptotic assumptions, especially in datasets with correlated predictors—an issue that is common in biomedical imaging, where multiple tumor shape and texture measures often covary strongly (Hosmer et al., 2013). When multicollinearity or limited sample sizes are present, the standard Wald-based confidence intervals and standard errors from maximum likelihood estimation may underestimate true variability, leading to overly optimistic inferences about predictor significance. These limitations have motivated the use of resampling-based methods, particularly the bootstrap, as a more flexible approach to quantifying uncertainty in logistic regression models.

The bootstrap, first introduced by Efron (1979), provides a nonparametric means of estimating sampling variability without relying on distributional assumptions. Its usefulness has been widely demonstrated in generalized linear models, where it can produce more accurate standard errors and confidence intervals than classical methods, especially in the presence of model misspecification or small-sample bias (Davison & Hinkley, 1997). Several variants of the bootstrap—including nonparametric, parametric, and residual-based versions—offer different strengths depending on the structure of the data and the model. For logistic regression specifically, the bootstrap has been shown to yield more reliable inference in high-dimensional or multicollinear settings by better capturing variability in coefficient estimates (Pittelkow & Sogaard, 2020).

In the context of breast cancer prediction, bootstrap methods have been used both to assess model stability and to evaluate the robustness of variable selection procedures. Studies applying the bootstrap to the WDBC dataset have found that only a subset of predictors consistently retain significance across resamples, highlighting the importance of stability analysis when identifying clinically meaningful features (Nehmzow & Smith, 2019). More broadly, the medical statistics literature emphasizes the role of resampling methods in validating classifier performance and

mitigating the risk of overfitting in clinical prediction models (Steyerberg et al., 2010). Together, this body of research supports the use of bootstrap-based uncertainty estimation as a rigorous complement to classical logistic regression inference in biomedical applications.

4. Method Description

4.1 Data and Variables

This analysis uses the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, which contains quantitative features extracted from digitized images of fine-needle aspirate (FNA) biopsies. Each observation describes a breast mass using several morphological characteristics such as radius, texture, perimeter, smoothness, concavity, compactness, symmetry, and fractal dimension. The binary outcome variable, diagnosis, indicates whether the tumor was malignant (1) or benign (0). Following common practice in the literature, this study focuses on the nine “mean” feature measurements, which summarize the average value of each characteristic across all detected nuclei within an image.

4.2 Logistic Regression Model

A logistic regression model was fit to predict tumor malignancy from the nine mean-based predictors:

$$\text{logit}(P(\text{malignant})) = \beta_0 + \beta_1(\text{mean_radius}) + \beta_2(\text{mean_texture}) + \dots$$

Maximum likelihood estimation was used to obtain the initial coefficient estimates, standard errors, and Wald confidence intervals using the `glm()` function in R with a binomial link. Due to the potential for multicollinearity among the morphological predictors, bootstrapping was implemented to evaluate the stability of the coefficient estimates and the reliability of the analytical inference.

4.3 Bootstrap Procedure

Three bootstrap schemes were implemented, each using 2,000 resamples:

4.3.1 Nonparametric Bootstrap

The nonparametric bootstrap resamples the original dataset with replacement, generating new datasets of the same size as the original. Logistic regression was fit to each resampled dataset, producing 2,000 bootstrap estimates for every coefficient. This approach makes minimal assumptions about the underlying data-generating mechanism.

4.3.2 Parametric Bootstrap

The parametric bootstrap assumes the fitted logistic regression model is correct. Using the estimated fitted probabilities from the original model, new binary outcomes were generated from a Bernoulli(p_i) distribution while holding the predictors fixed. Each simulated dataset was refit using logistic regression, yielding a parametric distribution of coefficient estimates.

4.3.3 Residual-Based Bootstrap

For the residual bootstrap, deviance residuals were extracted from the original model and resampled with replacement. Resampled residuals were added to the linear predictor to form new pseudo-linear predictors, which were then transformed back to probabilities using the inverse logit function. New binary responses were simulated from these adjusted probabilities, and the logistic model was refit. This approach preserves the relationship between predictors and the fitted values while allowing flexible resampling of unexplained variation.

4.4 Estimation of Uncertainty and Bias

For each bootstrap method, the empirical distribution of the 2,000 coefficient estimates was used to compute:

- Bootstrap standard errors (the standard deviation of bootstrap estimates)
- Bias estimates, defined as the difference between the mean bootstrap estimate and the original `glm()` coefficient
- Percentile-based 95% confidence intervals

The bootstrap results were then compared with the standard Wald inference from `glm()`, which provides analytical standard errors and confidence intervals under asymptotic assumptions.

4.5 Visualization and Comparative Evaluation

Kernel density plots were generated for each coefficient to visualize and compare the sampling distributions produced by the three bootstrap methods. These visualizations allowed assessment of coefficient stability, sign changes, multimodality, and the influence of multicollinearity. A comparison table was constructed to summarize the analytical and bootstrap standard errors and confidence intervals side by side. Together, these procedures provided a comprehensive evaluation of the reliability of logistic regression inference in the context of the breast cancer dataset.

5. Results

5.1 Logistic Regression Estimates

The logistic regression model was fit using the nine mean-based predictors from the WDBC dataset to classify tumors as malignant or benign. Several predictors showed large coefficient magnitudes, particularly those measured on very small scales such as `mean_smoothness`, `mean_concave_points`, and `mean_fractal_dimension`. However, the analytical Wald confidence intervals from `glm()` revealed that only a subset of predictors—primarily `mean_texture`, `mean_smoothness`, and `mean_concave_points`—had intervals that excluded zero, suggesting statistically meaningful associations with malignancy

5.2 Bootstrap Confidence Intervals

Bootstrap percentile confidence intervals were computed using nonparametric, parametric, and residual-based resampling (2,000 replications each). All three bootstrap methods produced consistent patterns of significance:

- Significant predictors across all methods:

`mean_texture`, `mean_smoothness`, `mean_concave_points`

- Non-significant or unstable predictors:

`mean_radius`, `mean_perimeter`, `mean_compactness`, `mean_concavity`, `mean_symmetry`, `mean_fractal_dimension`

For the significant predictors, the bootstrap intervals remained strictly positive under every resampling scheme, indicating stable and reproducible effects. In contrast, the

remaining predictors exhibited wide confidence intervals that overlapped zero, demonstrating substantial uncertainty and limited predictive contribution once multicollinearity among features was accounted for.

The close agreement across nonparametric, parametric, and residual bootstrap intervals suggests that the results are robust to assumptions about the data-generating mechanism.

5.3 Bootstrap Standard Errors and Bias

Bootstrap standard errors closely matched the analytical standard errors from `glm()`, with differences generally small across all coefficients. Predictors with small-scale measurements (e.g., `mean_smoothness`) exhibited larger standard errors due to the inherent scaling properties of the logistic coefficients, but this pattern was consistent across all bootstrap methods.

Bias estimates were also relatively small for most predictors. The largest biases were observed for `mean_fractal_dimension` and `mean_concavity`, both of which also displayed the widest bootstrap confidence intervals and the greatest instability across resamples. These patterns reinforce the conclusion that these predictors are weak and unreliable indicators of malignancy within the multivariable model.

Overall, bootstrap-based uncertainty estimation did not reveal hidden instability in the significant predictors or substantial underestimation of variability by the analytical model.

5.4 Visualization of Bootstrap Distributions

Density plots of the bootstrap coefficient distributions provided additional insight into coefficient stability. For `mean_texture`, `mean_smoothness`, and `mean_concave_points`, the bootstrap distributions were unimodal, tightly concentrated, and fully separated from zero. This visual evidence complements the confidence interval findings and confirms the robustness of these predictors.

In contrast, predictors such as `mean_radius`, `mean_perimeter`, and `mean_fractal_dimension` showed wide, diffuse, and sometimes skewed distributions with substantial overlap near zero. These patterns indicate that small changes in the data can produce large fluctuations in coefficient estimates, undermining their reliability in classification.

The similarity of the three bootstrap distributions within each panel further supports the conclusion that model results are stable and not sensitive to the choice of bootstrap method.

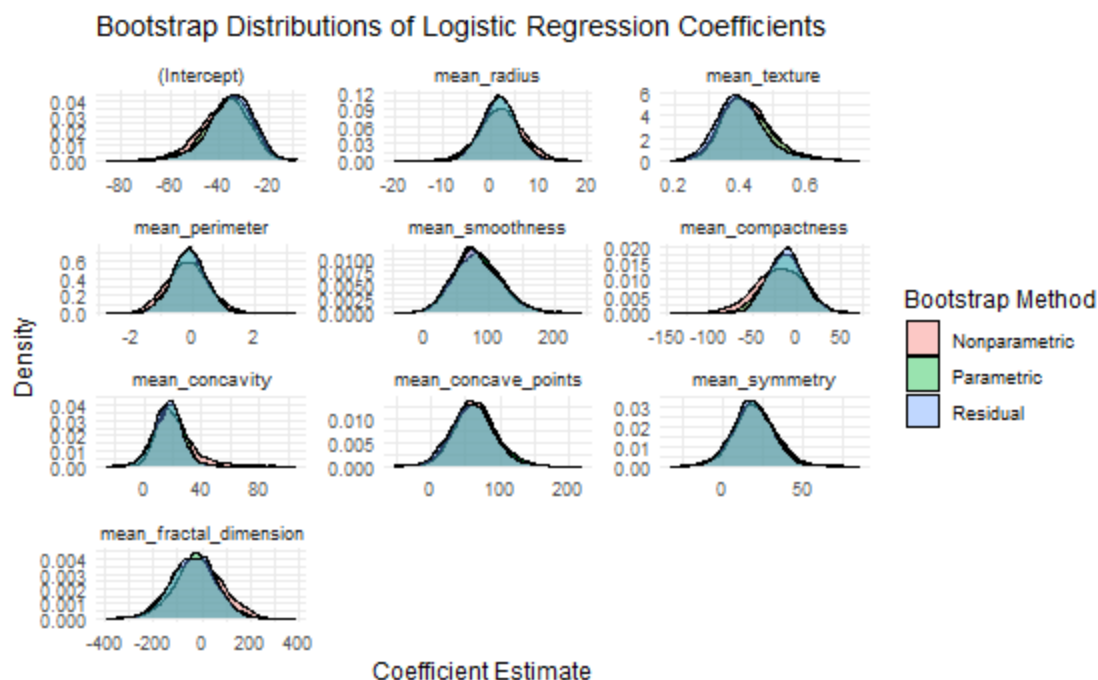


Figure 1. Bootstrap Distribution of Logistic Regression Coefficients

5.5 Comparison of Bootstrap and Analytical Inference

A comprehensive comparison table demonstrated that the bootstrap standard errors and confidence intervals closely aligned with the classical Wald estimates from `glm()`. Across all predictors, bootstrap intervals were similar in width and location to analytical intervals, offering no indication that the Wald method materially underestimated uncertainty. In cases where predictors were unstable (e.g., mean_fractal_dimension), both analytic and bootstrap methods consistently reflected high variability.

	Coefficient	Original	SE_glm	SE_Nonparametric	SE_Parametric	SE_Residual	Wald_Lower	Wald_Upper	Nonpar_Lower	Nonpar_Upper	Par_Lower	Par_Upper	Resid_Lower	Resid_Upper
(Intercept)	(Intercept)	-33.4222374	7.95820933	9.99946574	9.48285235	8.98270156	-49.0203277	-17.8241471	-60.0311293	-21.2442780	-57.1641006	-19.5547073	-54.8124540	-19.5405305
mean_radius	mean_radius	1.8249555	3.24414037	4.53528605	3.61269785	3.58568191	-4.5335596	8.1834706	-6.8375511	11.0637448	-5.2377155	8.7681474	-5.0291506	9.2235610
mean_texture	mean_texture	0.3828786	0.06307814	0.07380321	0.07731507	0.07363702	0.2592454	0.5065118	0.2940369	0.5844166	0.2873396	0.5967522	0.2728168	0.5715370
mean_perimeter	mean_perimeter	-0.1293387	0.50014295	0.69700630	0.55577634	0.55118191	-1.1096189	0.8509415	-1.4725667	1.2454756	-1.1924096	0.9616298	-1.2572678	0.9504834
mean_smoothness	mean_smoothness	73.6282088	34.02308270	36.31561608	37.81837494	37.06788173	6.9429667	140.3134509	16.7886822	159.8027471	14.3174003	164.8639568	12.4518841	157.6734098
mean_compactness	mean_compactness	-12.4900203	20.49390732	29.42693660	22.92547019	21.73865921	-52.6580787	27.6780380	-81.5893627	32.9202445	-59.5567039	31.1367946	-52.1907245	30.6648445
mean_concavity	mean_concavity	15.0911576	8.26194358	13.38635682	10.07652622	9.69211725	-1.1022518	31.2845670	2.0381184	56.1631295	-3.7232381	34.3212404	-3.5406248	35.2224030
mean_concave_points	mean_concave_points	59.9308134	28.75893258	29.21511868	31.51700538	30.26104349	3.5633055	116.2983212	8.2748787	123.2065992	4.7080592	129.6317906	2.1611099	120.5501231
mean_symmetry	mean_symmetry	18.1200747	10.93584129	13.29191266	12.52017469	12.15674363	-3.3141742	39.5543236	-6.5349098	47.2191005	-4.9214065	44.3323904	-4.3835003	42.5173062
mean_fractal_dimension	mean_fractal_dimension	-30.5830739	85.77589723	101.83620999	95.71089265	93.02394153	-198.7038325	137.5376846	-210.6881403	189.5441975	-224.2518186	154.9354602	-230.7887472	138.0532303

Figure 2. Wald vs Bootstrap Comparison Table

These results suggest that logistic regression's asymptotic approximations are adequate for this dataset and that bootstrapping primarily serves to validate the reliability of the analytical inference.

6. Discussion

The results of this analysis highlight the value of bootstrapping as a tool for assessing the stability and reliability of logistic regression models in biomedical classification tasks. Across all three bootstrap methods, only *mean_texture*, *mean_smoothness*, and *mean_concave_points* emerged as consistently significant predictors of malignancy, reinforcing prior findings that features reflecting nuclear irregularity and structural complexity are among the strongest indicators of cancerous tissue. The remaining predictors exhibited high uncertainty, wide confidence intervals, and substantial variability across resamples, suggesting that their apparent effects in the fitted model may be driven more by multicollinearity than by meaningful signal. Importantly, the bootstrap-based standard errors and confidence intervals closely matched the traditional Wald-based inferences from `glm()`, indicating that the asymptotic assumptions of logistic regression were adequate for this dataset and that analytical inference did not materially underestimate uncertainty. The strong agreement across nonparametric, parametric, and residual bootstrap procedures further suggests that the model is robust to reasonable assumptions about the data-generating process. Collectively, these findings demonstrate that while logistic regression can reliably identify the dominant predictors of malignancy, bootstrapping provides a valuable additional layer of validation—confirming which effects are reproducible, quantifying uncertainty more transparently, and helping distinguish stable biological signals from noise. This approach aligns with broader recommendations in the literature that emphasize the use of resampling methods to evaluate model reliability, especially in medical datasets characterized by correlated predictors and heterogeneous measurement scales.

7. Conclusion

This study applied logistic regression and three bootstrap resampling methods to evaluate the reliability of predictors used to classify breast tumors as malignant or benign in the Wisconsin Diagnostic Breast Cancer dataset. The bootstrap analyses confirmed that only a subset of features—specifically *mean_texture*, *mean_smoothness*, and *mean_concave_points*—provided stable and reproducible information about malignancy, while several commonly reported predictors demonstrated considerable uncertainty once the effects of correlated variables were accounted for. The close agreement between bootstrap-based uncertainty estimates and classical Wald confidence intervals indicates that the logistic regression model performed reliably for this dataset and that asymptotic inference was adequate despite the presence of multicollinearity. By revealing which predictors contribute true diagnostic signal and which are statistically unstable, the bootstrap results reinforce the importance of resampling-based validation in biomedical modeling. Overall, this analysis illustrates how combining logistic regression with bootstrap uncertainty estimation can strengthen the interpretability and robustness of predictive models, offering a practical framework for future clinical and statistical applications involving complex, high-dimensional medical data.

8. References

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9. Appendix

```
# Read data
library(readxl)

bc <- read_excel("C:/Users/stewa/OneDrive/Documents/Breast Cancer.xlsx")
head(bc)

set.seed(123)

# Model formula you want to study:
form <- diagnosis ~ mean_radius + mean_texture + mean_perimeter +
  mean_smoothness + mean_compactness + mean_concavity +
  mean_concave_points + mean_symmetry +
  mean_fractal_dimension

n <- nrow(bc)

#Non parametric
library(boot)

# Statistic function for boot()
logit_stat <- function(data, indices) {
  d <- data[indices, ]
  fit <- glm(form, data = d, family = binomial)
  coef(fit)
}

B <- 2000

boot_nonpar <- boot(data = bc, statistic = logit_stat, R = B)

nonpar_ci <- t(apply(boot_nonpar$t, 2, quantile, probs = c(0.025, 0.975)))
nonpar_ci

#Parametric
# Fit model once to get fitted probabilities
fit0 <- glm(form, data = bc, family = binomial)
p_hat <- fitted(fit0)

B <- 2000
coefs_par <- matrix(NA, nrow = B, ncol = length(coef(fit0)))
colnames(coefs_par) <- names(coef(fit0))

for (b in 1:B) {
  # Simulate new responses from Bernoulli(p_hat)
  y_star <- rbinom(n = n, size = 1, prob = p_hat)

  bc_star <- bc
  bc_star$diagnosis <- y_star

  fit_star <- glm(form, data = bc_star, family = binomial)
  coefs_par[b, ] <- coef(fit_star)
}
```

```

# 95% percentile CIs for parametric bootstrap
par_ci <- t(apply(coefs_par, 2, quantile, probs = c(0.025, 0.975)))
par_ci

#Residual
# From previous fit
eta_hat <- predict(fit0, type = "link")
res_dev <- residuals(fit0, type = "deviance")

B <- 2000
coefs_resid <- matrix(NA, nrow = B, ncol = length(coef(fit0)))
colnames(coefs_resid) <- names(coef(fit0))

for (b in 1:B) {
  res_star <- sample(res_dev, size = n, replace = TRUE)

  # New linear predictor and probabilities
  eta_star <- eta_hat + res_star
  p_star <- plogis(eta_star)

  # Simulate new responses and refit
  y_star <- rbinom(n = n, size = 1, prob = p_star)

  bc_star <- bc
  bc_star$diagnosis <- y_star

  fit_star <- glm(form, data = bc_star, family = binomial)
  coefs_resid[b, ] <- coef(fit_star)
}

# 95% percentile CIs for residual-based bootstrap
resid_ci <- t(apply(coefs_resid, 2, quantile, probs = c(0.025, 0.975)))
resid_ci

#Nonpar SE & Bias
fit0 <- glm(form, data = bc, family = binomial)
orig_coef <- coef(fit0)
orig_coef

# Standard errors (SD of bootstrap estimates)
se_nonpar <- apply(boot_nonpar$t, 2, sd)
se_nonpar

# Bias = mean(bootstrap estimates) - original coefficient
bias_nonpar <- colMeans(boot_nonpar$t) - orig_coef
bias_nonpar

#Para SE & Bias
se_par <- apply(coefs_par, 2, sd)
se_par

bias_par <- colMeans(coefs_par) - orig_coef
bias_par

```

```

#Residual SE & Bias
se_resid <- apply(coefs_resid, 2, sd)
se_resid

bias_resid <- colMeans(coefs_resid) - orig_coef
bias_resid

#Bootstrap Distribution Plots
library(dplyr)
library(tidyr)
library(ggplot2)

coef_names <- names(orig_coef)

# Convert each bootstrap result to a data frame and tag with method
df_nonpar <- as.data.frame(boot_nonpar$t)
colnames(df_nonpar) <- coef_names
df_nonpar$Method <- "Nonparametric"

df_par <- as.data.frame(coefs_par)
colnames(df_par) <- coef_names
df_par$Method <- "Parametric"

df_resid <- as.data.frame(coefs_resid)
colnames(df_resid) <- coef_names
df_resid$Method <- "Residual"

# Stack them into one long data frame
boot_long <- bind_rows(df_nonpar, df_par, df_resid) |>
  pivot_longer(
    cols = all_of(coef_names),
    names_to = "Coefficient",
    values_to = "Estimate"
  )

boot_long$Coefficient <- factor(
  boot_long$Coefficient,
  levels = coef_names
)

# Density plots of bootstrap coefficients
ggplot(boot_long, aes(x = Estimate, fill = Method)) +
  geom_density(alpha = 0.4) +
  facet_wrap(~ Coefficient, scales = "free", ncol = 3) +
  theme_minimal() +
  labs(
    title = "Bootstrap Distributions of Logistic Regression Coefficients",
    x = "Coefficient Estimate",
    y = "Density",
    fill = "Bootstrap Method"
  )

```

```

#Wald Comparison
glm_sum <- summary(fit0)$coefficients
se_glm <- glm_sum[, "Std. Error"]

# 95% Wald CI: beta_hat ± 1.96 * SE
wald_ci_low <- orig_coef - 1.96 * se_glm
wald_ci_high <- orig_coef + 1.96 * se_glm

# Nonparametric percentile CIs
ci_nonpar <- apply(boot_nonpar$t, 2, quantile, probs = c(0.025, 0.975))

# Parametric percentile CIs
ci_par <- apply(coefs_par, 2, quantile, probs = c(0.025, 0.975))

# Residual percentile CIs
ci_resid <- apply(coefs_resid, 2, quantile, probs = c(0.025, 0.975))

bootstrap_comparison <- data.frame(
  Coefficient = coef_names,
  Original = orig_coef,
  SE_glm = se_glm,
  SE_Nonparametric = se_nonpar,
  SE_Parametric = se_par,
  SE_Residual = se_resid,
  Wald_Lower = wald_ci_low,
  Wald_Upper = wald_ci_high,
  Nonpar_Lower = ci_nonpar[1, ],
  Nonpar_Upper = ci_nonpar[2, ],
  Par_Lower = ci_par[1, ],
  Par_Upper = ci_par[2, ],
  Resid_Lower = ci_resid[1, ],
  Resid_Upper = ci_resid[2, ]
)

View(bootstrap_comparison)

```